Problem (1)

The two balls meet at the same time at $y = h/2$.

Ball A: $h/2 = h - \frac{1}{2}gt^2$ (1) $\rightarrow t = \sqrt{h/g}$ (2)

Ball B: $h/2 = 0 + v_{oB}t - \frac{1}{2}gt^2$ (3)

Subtract (3)-(1) $h = v_{oB}t$ Substitute from (2) $\rightarrow v_{oB} = \sqrt{gh}$

Problem (2)

a) For the two ball to collide, both should have the same x and y coordinates at the time of collision. During the projectile motion $x_A = 25\,t$ and $x_B = 35\,t$, so $x_A \neq x_B$ at anytime t. Hence, the two balls will not collide (no need to check the y distance)

b) Time calculations are as follows.

Ball A: $\theta = \tan^{-1} 15/25 = 31°$, $v_{oA}^2 = 25^2 + 15^2 = 825$, $x_{maxA} = \frac{v_{oA}^2 \sin 2\theta}{g} = 76.6\ m$

$\rightarrow t_A = \frac{x_{maxA}}{v_{oxA}} = 3.06\ s$

Ball B: $y_B = y_{oB} + v_{oyB}t_B - \frac{1}{2}gt_B^2$ $\rightarrow 0 = 20 - \frac{1}{2}(9.8)t_B^2$, $\rightarrow t_B = 2.02\ s$

c) Distances in the x directions are as follows.

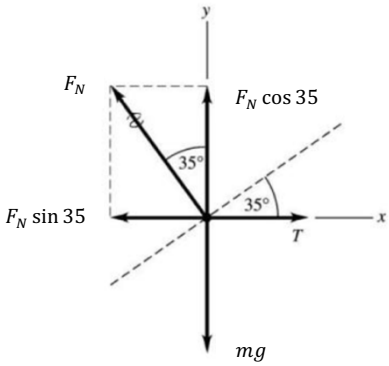
From part (b), $x_{maxA} = 76.6\ m$

And $x_{maxB} = v_{oxB}t_B = 35(2.02) = 70.7\ m$

$\rightarrow x_{maxA} - x_{maxB} = 76.6 - 70.7 = 5.9\ m$

Problem (3)

a) The free-body diagram is as shown.



b) The surface pushes with the normal force.

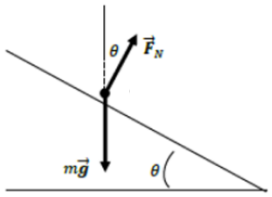
$F_N \cos 35 = mg$ $\rightarrow F_N = \frac{mg}{\cos 35} = 1.22\ mg$

c) The tension is give by,

$T = F_N \sin 35 = 0.7\ mg$

Problem (4)

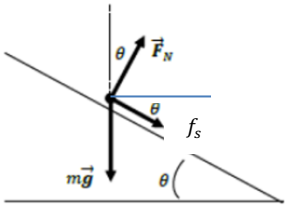
a) With no friction, the free-body diagram is as shown. In this case, the maximum velocity without skidding is found as follows.



$F_N \cos\theta = mg$ (1) $F_N \sin\theta = \frac{mv^2}{R}$ (2)

Divide (2)/(1): $\tan\theta = \frac{v^2}{Rg}$ $\rightarrow v_{\max_no_f} = \sqrt{Rg\tan\theta}$

b) With friction, the free-body diagram is as shown. In this case, the maxium velocity with no skidding is found as follows.



$F_N \sin\theta + \mu_s F_N \cos\theta = \frac{mv^2}{R}$ (1) $F_N \cos\theta - \mu_s F_N \sin\theta = mg$ (2)

Divide (1)/(2): $v^2 = Rg\left(\frac{\sin\theta + \mu_s \cos\theta}{\cos\theta - \mu_s \sin\theta}\right)$ $\rightarrow v_{\max_with_f} = \sqrt{Rg\left(\frac{\sin\theta + \mu_s \cos\theta}{\cos\theta - \mu_s \sin\theta}\right)}$

Divide by $\cos\theta$, $v_{\max_with_f} = \sqrt{Rg\left(\frac{\tan\theta + \mu_s}{1 - \mu_s \tan\theta}\right)} = \sqrt{Rg\left(\frac{v_{\max_no_f}^2 + \mu_s Rg}{Rg - \mu_s v_{\max_no_f}^2}\right)}$